



Bank Marketing

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Agenda

What this report covers

01 Introduction

02 Dataset Overview/
Preprocessing

03 Modeling Approach

04 Results

05 Insights

06 Recommendations

07 Conclusion

08 Q&A

Business Problem

Higher number of target customers saying 'No'
Targeting the wrong audience

Goal

Determine Machine Learning method best for predictions in successful telemarketing campaigns leading to subscriptions

Why does this analysis matter?

- Reduces inefficiency in time, labor, and budget spending
- Provides understanding to allocate correct target market
- Increases subscription Rates



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Our Data / Preprocessing

Column Removal

Selection

- Removing Incomplete Values
- Binning Values
- Making dummy variables

Classifying Data

Two Classifiers

Decision Tree
vs.
Logistic Regression

Data Modification

Experiment

- What parameters need to be turned for greater accuracy?

Modeling Approach

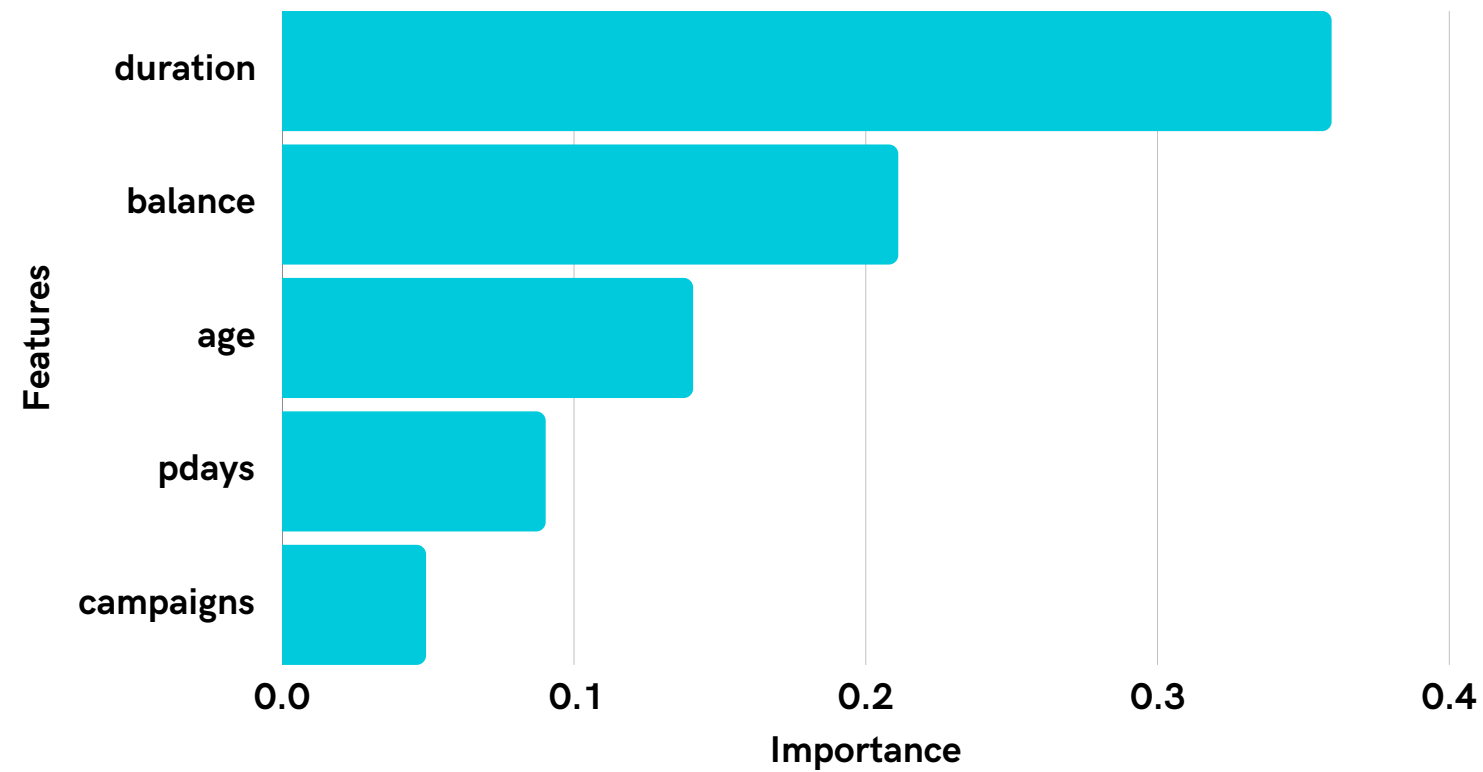
Why Decision Tree and Logistic Regression Models?

Decision Trees and Logistic Models are often standard for determining classification models, and have different attributes that make them better for different data sets. Also allows for tuning via pruning and variable weightage

Our models did well for no churn evaluations but not for churn evaluations



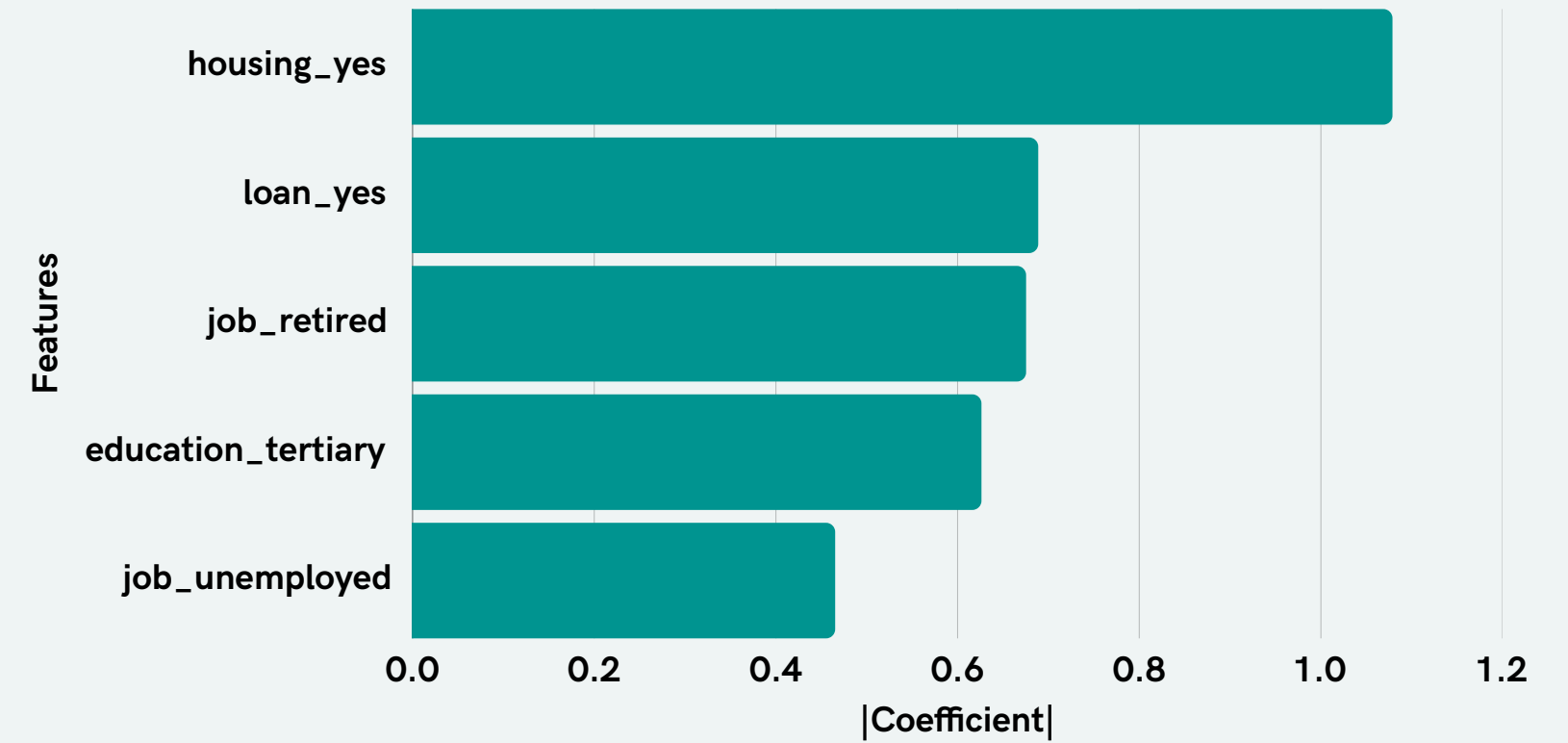
Feature Importances for Decision Tree Classifier



Decision Tree

10 fold avg f-1 score:
0.3407

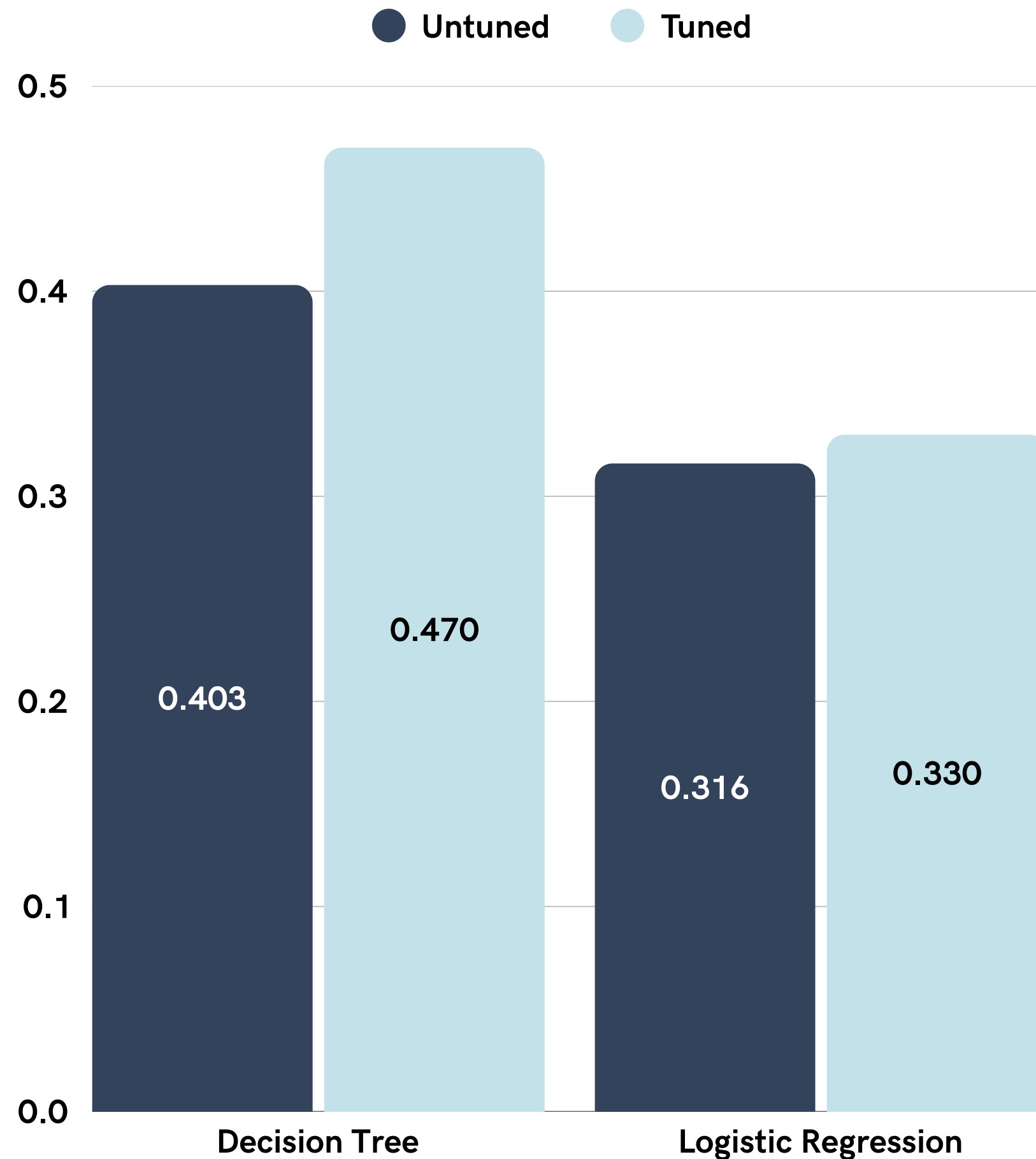
Feature Importances (Absolute Coefficients) for Logistic Regression Classifier



Logistic Regression

10 fold avg f-1 score:
0.2958

Logistic Regression vs. Decision Tree F1



Results

Tuning for Hyperparameters

In both Tuned and Untuned Data, Decision Trees worked better for our data

- Untuned: good still misses some positives
- Tuned: better balance; more positives (+)
Better for generalizations
- Evaluated using F1 score to measure real world probability
- Tuned Decision Tree → best precision and recall (best choice from all)

Insights

As of December 1, 2025

Results Breakdown

Decision Tree Preferred

DT: balanced and reliable

How well we identify 'Yes' customers = 0.470

Removed N/A & leakages

Campaign Efficiency

Optimize Contact Strategies

More contacting → lower chance of 'Yes'

Successful past interactions → higher chance of 'Yes'

Customer Prioritization

Top Predictors

Retired

Unemployed

Loan Free

Previous Campaign Contacts

Recommendations



Prioritize Customer Segments

Find similar characteristics; Create own group focus

Retired, Higher Level Education, No Loans, Previous Campaigns



Tailor Messaging to Customers

Retired → security

Higher education → long term growth

Repeated contacting → avoid customers; no longer contact



Prioritize Decision Tree as Preferred Model

Produced highest real world performances simplest explanations

Conclusion

Decision Tree

Best model

Strongest real world predictive analysis without overfitting

Key Models

What stands out

Job type, loan status, education level, past campaign

Predictive Modeling

(+) For Marketing Efficiency

Turns effort to higher likely candidates

Thank you!

Feel free to reach out to us if you have any questions.



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